Diabetes App Challenge · Feature strategy for Mustafa

# DeepSleep — Selected Feature Set

A sleep tab + softer, glucose-aware food agent, built as one closed behavioral loop.

Prepared 18 July 2026 · Selected sections: A, B, C · Section D cached for Phase 2 · Grounded in the *Deep Sleep and Eye-First Diabetes App Strategy* report and the existing Flutter codebase.

**The core thesis.** This is not a generic sleep tracker or meal planner. It is a **closed behavioral loop**: use the user's overnight Libre glucose to infer their **overnight glucose phenotype** → have the **food agent** generate earlier, more sleep-compatible evening meals from what is actually in the fridge → use a **sleep layer** focused on regularity and restoration to keep the whole system honest. Every recommendation is **soft and CGM-conditional** — the app informs, it never dictates. The app already imports overnight LibreView data and already runs a Gemini food agent, so most of this reuses code that exists.

**Recommended build order for the challenge:** S1 → S2 → S3/S4 → S8 → S7 → S10, then layer the food-agent changes F1/F2/F3 → F4 on top. This delivers the full closed loop the evidence argues wins, and nearly all of it reuses the existing CGM import, meal logging, and food-agent code.

PRIORITY: HIGHEST · PRIORITY: MEDIUM · PRIORITY: LOWER · EFFORT: LOW · EFFORT: MEDIUM · EFFORT: HIGH

— Priority follows the evidence report; effort is estimated for this app.

## A · Sleep Tab — the core loop

The backbone: log sleep, read the overnight glucose story from data you already have, and nudge the two biggest levers — dinner timing and sleep regularity.

### S1 Sleep log (manual)

HIGHEST · LOW

Bedtime, wake time, quality (1–5), and number of wake-ups. The foundation everything else reads from.

**Reuses:** new Firestore collection under `users/{uid}/sleepLogs`, mirroring the existing meals/workouts repository pattern.

### S2 Overnight glucose phenotype engine

HIGHEST · MEDIUM

The report's #1 feature. For each logged night, read the CGM values between bed and wake and classify the night: **late-dinner hyperglycemia**, **downward-trending pre-bed**, **high overnight variability**, or **frequent lows**. Built from standard consensus metrics — TIR, TBR <70 & <54, TAR, bedtime trend, overnight CV.

**Reuses:** `GlucoseReading` + `GlucoseRepository`; thresholds already in `GlucoseSettings` (low 70 / high 180).

### S3 Dinner timing coach

HIGHEST · LOW–MED

The report treats *when* dinner happens as a first-class variable and the most scalable metabolic lever. Track the dinner-to-bed gap and gently nudge toward finishing ~3–4h before bed — **only** when a user's own overnight highs actually follow late meals.

**Reuses:** existing meal-log timestamps + S1 bedtime; surfaces alongside S2.

### S4 Sleep regularity & the 7–9h sweet spot

HIGHEST LOW

Sleep *timing variability* hurts time-in-range as much as duration does. Track bed/wake consistency, surface the 7–9h target plainly, and offer an optional bedtime reminder. Reward consistency, don't punish one-off nights.

**Reuses:** S1 history; optional local notification.

### S5 "Last night → this morning" card

MEDIUM LOW

Ties sleep duration/regularity to next-morning fasting glucose and the day's first spike. Over time: "*Your steadiest glucose days follow 7h+ nights with a consistent bedtime.*"

**Reuses:** S1 + morning `GlucoseReading` s.

### S6 Deep-sleep (N3) via wearable / Health Connect

LOWER HIGH

**Phase 2, optional.** Real sleep stages including N3 from a wearable / Apple Health / Google Health Connect. The report explicitly cautions to stay **modest** about N3 claims — consumer devices are for trend tracking, not physiologic proof. The manual log + overnight CGM (S1–S4) is the powerful, buildable core; this is an add-on.

**Reuses:** same "import external data" pattern as the LibreView CSV importer.

## B · Sleep Tab — safety & insight

The overnight-risk layer. This is where a combined Libre + sleep + meal app beats a generic sleep tracker — it can tell "disrupted by biochemistry" apart from "disrupted by hygiene/timing."

### S7 Conditional bedtime-snack advisor

HIGHEST MEDIUM

Report-Highest, but deliberately narrow. Suggest a small protein-forward snack **only** when the pattern is low-normal pre-bed glucose + a downward trend (e.g. after evening exercise) — **never routine**, since reflexive late snacking worsens overnight glycemia. Always framed as optional with a "check with your clinician" note.

**Reuses:** S2 phenotype + S8 low detection; a small rules engine, not autonomous dosing.

### S8 Nocturnal-low flag

HIGHEST LOW

Detects overnight readings <70 and <54 mg/dL from CGM data and surfaces them gently — a genuine safety issue and a sleep disruptor in its own right. Feeds S7.

**Reuses:** `GlucoseReading` stream filtered to the sleep window.

### S9 Quiet-overnight / tiered alerts

MEDIUM LOW

The report warns that alarm fatigue fragments sleep. Design principle: evening nudges are fine, overnight stays quiet by default, escalate only for meaningful predicted risk. Let the user choose whether the sleep layer focuses on *regularity*, *overnight lows*, or *late-dinner highs*.

**Reuses:** notification settings; governs how S3/S4/S7/S8 speak.

### S10 Sleep summary on the Home dashboard

MEDIUM LOW

A compact "Last night" card on the existing Today dashboard, sitting beside Activity and Glucose so sleep isn't siloed.

**Reuses:** `TodayDashboard` card layout, next to `GlucoseContextCard`.

## C · Food Agent changes — soft, glucose-aware

Your "recommendations, not hard sets." The agent still returns exactly what the user asks for — it just orders it more helpfully and adds optional, evidence-based context. All changes center on `GeminiMealService` and `MealRecommendation`.

### F1 Evening dual-score ranking

HIGHEST LOW

For dinner/late meals, bias toward **fiber + protein + unsaturated fat** (legumes, yogurt, fish, tofu, oats, veg, olive oil) and gently down-rank late refined carbs and saturated-fat-heavy plates. It still returns the requested dish — it just orders and annotates.

**Reuses:** prompt change in `GeminiMealService._buildPrompt`.

### F2 Soft "heads-up" note per dish

HIGHEST LOW

Add an optional `glucoseNote` / `sleepNote` field: e.g. "*Carb-forward for this close to bed — pairing with the yogurt in your fridge softens the overnight rise.*" Shown as an info chip, never a warning.

**Reuses:** new field on `MealRecommendation` + a chip in `MealRecommendationCard`.

### F3 Time-of-day awareness

HIGHEST LOW

The agent already knows `MealType`; make evening meals lightly favor lighter/earlier-friendly options **without refusing** the request. Pairs with the dinner-timing coach (S3).

**Reuses:** existing `MealType` in `AgentRequest`.

### F4 Personalized from the user's phenotype

HIGHEST MEDIUM

Feed the S2 night-pattern into the agent: "*Rice tends to spike you overnight — here's a lower-GL swap, but your pick is still here.*" This is the report's "adapt the plan, don't prescribe it."

**Reuses:** S2 output threaded into `AgentRequest`.

### F5 "Make it more balanced?" opt-in

MEDIUM LOW

One tap keeps the user's dish but returns a **small** tweak (a protein/fiber pairing, a smaller carb portion) instead of replacing it. Consent-based, never forced.

**Reuses:** the existing "Suggest Something Else" feedback flow.

### F6 Softer system-prompt framing

MEDIUM LOW

Retune from "recommend healthy dishes" to "honor what they want, then *offer* optional insight." Cheap change, big tone shift toward the soft-recommendation philosophy.

**Reuses:** prompt text in `GeminiMealService._buildPrompt`.

### F7 Learn from observed overnight response

LOWER HIGH

The report's reinforcement-learning idea. Over time, re-rank ingredients by how *this user's* overnight glucose actually responded, not just nutrient estimates.

**Reuses:** Meal Lab correlation data already being logged.

CACHED · PHASE 2 · DO NOT BUILD NOW

## D · Eye-health & clinical credibility

Recorded for later per your decision. Nothing in this section is scheduled for implementation. The report is emphatic that any eye work stays as education & screening adherence — never diagnosis.

### E1 Eye-health education + screening workflow

MEDIUM LOW–MED

Annual dilated-exam reminders, vision-change red-flags, dry-eye and OSA prompts. Education and screening adherence only.

### E2 Weekly clinician / coach summary

MEDIUM MEDIUM

Exportable AGP + sleep-regularity + dinner-time report. Adds real-care credibility.

### E3 "Support, not dosing" guardrails

MEDIUM LOW

General-wellness positioning, no insulin dosing, consent-separated data. A stance to bake in when the time comes.

**Open decision to confirm before planning:** the recommended core is **manual sleep log + your existing overnight CGM data** (ships for the challenge; S1–S5, S7–S10). **S6 (wearable / Health Connect N3)** is included in Section A but flagged as Phase 2 — confirm whether you want it in the first build or deferred alongside D.

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**Evidence basis.** Derived from *Deep Sleep and Eye-First Diabetes App Strategy for Mustafa*, which synthesizes CGM–sleep association studies (7–9h sweet spot; sleep-timing variability vs. time-in-range), meal-timing and glycemic-index RCTs, a bedtime-snack trial and the 2025 DailyDose smart-snack study, and Abbott/FDA documentation on LibreView's intended use. Priorities in this document mirror that report; effort estimates are specific to the current Flutter + Firebase + Gemini codebase. All recommendations are behavioral and educational — no autonomous insulin dosing or retinal diagnosis.